

Two errors in the network model's load and PV equations

What they were, why nobody saw them,
and what they change

Tulio Coura · 26 August 2026 · all nine transformers

The short version. The model contained two mistakes in the equations that describe how much current a house draws when its voltage changes. Both had been there since the beginning. Both are now fixed, and every transformer has been re-run.

The effect is stranger than you might expect. The flexibility region does not grow or shrink so much as *turn*: it reaches further into reactive import and less far into export, by up to 33.7 kW. That rotation appears on **all nine transformers**, in the same direction, which is what you would predict from the specific terms that were wrong.

Where the battery schedule is unchanged, the region's *area* moves by a median of 0.05%. Where the fix also moved the schedule, it moves by up to 7%. So conclusions about *how much* flexibility a feeder has are safe; conclusions about *specific operating points*, especially reactive ones, are the ones to re-examine.

1. What these equations are for

The physics of a power network is nonlinear. A house that draws a fixed amount of power draws *more current* when its voltage sags, and less when it rises. That relationship is a division, not a multiplication, and an optimiser cannot work with it directly.

So the model does what engineers always do with an awkward curve: it replaces it with a straight line. It picks a starting voltage — here, the flat-start value where every bus sits at its nominal voltage — and asks a simple question at that point:

if the voltage moves a little, how much does the current move?

The answer is a set of slopes. There are eight of them: one for each combination of {real, imaginary} current against {real, imaginary} voltage, for loads and for PV. They are ordinary partial derivatives, worked out once by hand and typed into the model as formulas.

Two of those eight were wrong. Not the physics around them — the network equations, the limits, the battery model are all sound. Just two of the eight slopes.

2. The first error: a parenthesis in the wrong place

The imaginary part of a load's current is

$$I_{\text{im}} = \frac{P_0 V_{\text{im}} - Q_0 V_{\text{re}}}{V_{\text{re}}^2 + V_{\text{im}}^2}.$$

Differentiating with respect to V_{im} gives two terms — one carrying the active power P_0 , one carrying the reactive power Q_0 :

$$\frac{\partial I_{\text{im}}}{\partial V_{\text{im}}} = \frac{P_0(V_{\text{re}}^2 - V_{\text{im}}^2) + 2Q_0 V_{\text{re}} V_{\text{im}}}{(V_{\text{re}}^2 + V_{\text{im}}^2)^2}.$$

In the model, the closing bracket sat one term too late, so the Q_0 term ended up *inside* the multiplication by P_0 :

```

was    P0*( Vre0^2 - Vim0^2 + 2*Q0*Vre0*Vim0 ) / (.)^2
now    ( P0*(Vre0^2 - Vim0^2) + 2*Q0*Vre0*Vim0 ) / (.)^2

```

Why this is worse than a typo

You might expect a stray bracket to distort a term. This one removed it.

Everything inside the model is in per unit on a 1 MW base. A one-kilowatt household load is therefore 0.001. Multiplying the reactive term by P_0 did not tilt it — it shrank it by a factor of about a thousand.

In practice, the reactive coupling in that equation was simply **missing**.

3. The second error: the wrong quantity

The real part of a PV system's current is

$$I_{\text{re}}^{\text{PV}} = \frac{G V_{\text{re}}}{V_{\text{re}}^2 + V_{\text{im}}^2}, \quad \text{so} \quad \frac{\partial I_{\text{re}}^{\text{PV}}}{\partial V_{\text{im}}} = \frac{-2 G V_{\text{re}} V_{\text{im}}}{(V_{\text{re}}^2 + V_{\text{im}}^2)^2},$$

where G is the PV output. The model used *net* demand — the house load minus its PV, `P0minGenP` — where it should have used the PV output alone:

```

was    ( -2*P0minGenP(d,t)*Vre0*Vim0 ) / (.)^2
now    ( -2*GenP0(d,t)*Vre0*Vim0 ) / (.)^2

```

Every other PV slope in the model uses the PV output. The current being differentiated is defined from the PV output. Only this one line disagreed.

Why this one changes sign with the sun

Write out what the model used against what it should have used, in per unit:

Time of day	used	correct	effect
Midday, PV 3 kW, load 1 kW	−0.0020	+0.0030	sign flipped
Dawn or dusk, PV 0.5 kW, load 1 kW	+0.0005	+0.0005	accidentally right
Night, PV 0 kW, load 1 kW	+0.0010	0	invented from nothing

At midday, when PV dominates the house load, the term pointed **the wrong way**. At dawn and dusk, where load and PV happen to be similar, subtracting one from the other lands close to the right answer by luck. At night it produced a PV sensitivity for a panel generating nothing.

This is a testable prediction: the damage should be worst around noon and near zero at either end of the day. That is exactly what the measurement shows.

A check that the fix is right

After correction, this slope is character-for-character identical to another one in the model that was always correct — the one for imaginary current against real voltage.

That is not a coincidence and not a copy-and-paste. For PV, the two cross-slopes are mathematically the same quantity:

$$\frac{\partial I_{\text{re}}}{\partial V_{\text{im}}} = \frac{\partial I_{\text{im}}}{\partial V_{\text{re}}} = \frac{-2 G V_{\text{re}} V_{\text{im}}}{(V_{\text{re}}^2 + V_{\text{im}}^2)^2}.$$

The corrected line landed on a value the model already contained, independently, in another place. The remaining six slopes were re-derived by hand and are all correct.

4. Why nobody saw them

This is the uncomfortable part: **neither error makes anything fail.**

These slopes are not constraints. Nothing can violate them. They do not affect whether a solution exists, and they do not change what the solver reports. Both the old and the new model return *Optimal* on all 5,184 problems — 576 on each of the nine transformers — report healthy voltages, and pass every after-the-fact physical check we have: no voltage violations, in either version, anywhere.

The only symptom is a slightly wrong sensitivity, which flows quietly into the power balance and out into the results.

There was nothing to notice. The errors were found by re-deriving all eight slopes from scratch during a full read of the model — not by chasing a bad number.

5. What was measured

Every transformer was run twice, before and after the fix, with everything else held identical: the per-transformer reduced network, the exact formulation, 48 half-hours, 12 directions each. Nine transformers, 5,184 boundary points per pass.

Solve status	576 of 576 optimal, every transformer, both passes		
Directions that changed	5,180 of 5,184		
Largest boundary movement	33.7 kW (TR5)		
Typical boundary movement	1.8 kW		
Voltage violations introduced	none, on any transformer		
Change in runtime	negligible		

Nearly every direction moved. Yet the area barely did. Those two facts only fit together one way: the region is **rotating**, not growing or shrinking.

Reactive-import side		Export side	
60°	+3.5 kW	180°	−2.9 kW
90°	+3.6 kW	210°	−3.5 kW
120°	+1.5 kW	240°	−2.3 kW

One side pushes out, the other pulls in, by matching amounts. The area survives because the two cancel.

And this is not one transformer’s quirk. All nine push out on the reactive-import side and pull in on the export side — nine out of nine, same sign, same shape. That is the strongest evidence that what is being seen is the correction doing what the algebra says it should, rather than numerical noise.

This is what you would predict from the two errors. Both are *cross* terms — they link the real and imaginary parts of the electrical quantities, which is another way of saying they govern the interaction between active and reactive power. Correcting them moves reactive power. It is not random error; it is a systematic shift along one axis, and it does not average out.

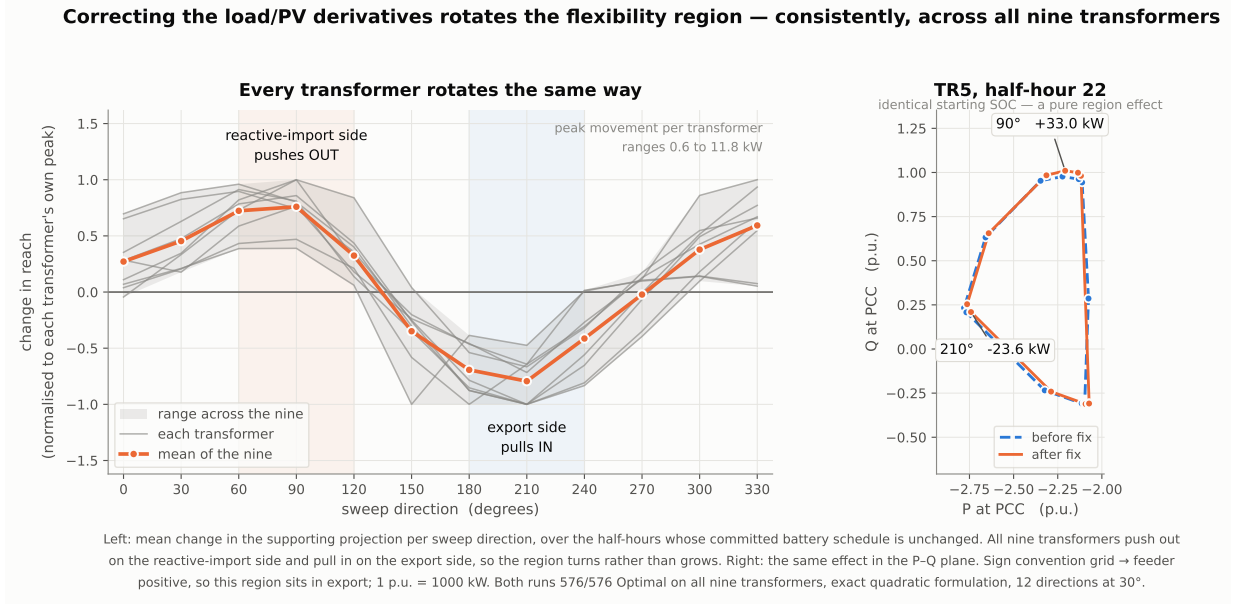


Figure 1: Left: the change in reach by sweep direction, one grey line per transformer, each normalised to its own peak so the shapes can be compared. Every one of the nine pushes out on the reactive-import side and pulls in on the export side. Right: the same effect in the P-Q plane, on a half-hour where the battery schedule is unchanged, so the movement is the region and nothing else. The right-hand edge, where active power sets the limit, hardly moves.

One honest caveat

The battery schedule also changed.

The model decides what the batteries actually do using the same equations, so correcting them moved that decision too. In those half-hours the difference between the two runs mixes two things: the region moving, and the batteries starting somewhere else. Splitting them apart matters, because the two give very different numbers:

Half-hours	count	median area change	worst
Battery schedule unchanged	151	-0.05%	-2.8%
Battery schedule also moved	281	-0.26%	-7.1%

So the region on its own is stable — 146 of the 151 clean half-hours move by less than 1%. The large swings live where the fix also rescheduled the batteries.

Neither number is wrong. The new schedule is the correct one; the old was produced by the faulty equations. But it does mean the published battery trajectories are affected too, not only the regions.

That rescheduling is narrow. On the worst transformer it touches **3 batteries out of 77**, in the midday window, and the worst of them charges to a full state of charge where the old model left it at 60%. By the end of the day 75 of those 77 batteries are back to identical. The fix restores the PV coupling term that was sign-flipped at exactly those hours, so midday is precisely where one would expect it to bite.

6. What this means

If a conclusion is about how much flexibility a feeder can offer, it stands. Where the battery schedule is unchanged the area moves by a median of 0.05%, and 146 of 151 such half-hours move by less than 1%.

If a conclusion is about a specific operating point — a stated maximum import or export at a given hour — it moves by up to 33.7 kW.

If a conclusion is about reactive power, that is where to look hardest. The error is systematic on that axis, it points the same way on all nine transformers, and it does not average out.

If a result depends on the battery trajectory rather than the region, check the midday window. Seven of the nine transformers reschedule there.

The two lines sit in the network physics that the rest of the model was built on. Anything else that model has been used for carries them too.

7. What happens next

The fix is committed and all nine transformers have been re-run and measured. Nothing further is pending on this correction.

One larger approximation remains untouched. The model draws its straight-line approximation once, at the flat-start voltage, and never redraws it as the solution moves away. The gap between the approximation and the true voltage reaches about 1.6% — larger than the error just corrected. Fixing that would mean regenerating everything again, so it is worth deciding deliberately rather than drifting into it.